

FREE GROUPS ARE L^2 -SUBGROUP RIGID

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ABSTRACT. In this paper, we introduce the notion of L^2 -subgroup rigid groups and demonstrate that free groups are L^2 -subgroup rigid. As a consequence, we establish the equivalence between compressibility, inertness, strong inertness, and L^2 -independence for a finitely generated subgroup of a free group, confirming a conjecture by Dicks and Ventura as well as the one by Antolin and Jaikin-Zapirain.

1. INTRODUCTION

The study of finitely generated subgroups of a free group is a classical topic in combinatorial group theory. The standard approach employs Stallings' graph theory techniques introduced in [Sta83] (see also [KM02]). However, it has been realized that these techniques have limitations in addressing problems related to the intersections of subgroups, as exemplified by the case of the Hanna Neumann Conjecture. The proofs of the conjecture by Friedman[Fri15] and Mineyev[Min12], as well as a more recent proof by Jaikin-Zapirain[JZ17] use different algebraic tools. These ideas were further developed by Antolin and Jaikin-Zapirain [AJZ22], where the notion of L^2 -independence was introduced. This led to the proof of a generalized version of the Hanna Neumann conjecture for various families of groups [BK23, FM23]. In this paper, we introduce the notion of L^2 -subgroup rigid group, providing another powerful tool for proving new results on subgroups of groups possessing this property.

Let G be a group and let $\mathcal{U}(G)$ denote the ring of affiliated operators of G (see Section 2.4 for the definitions). For a left $\mathbb{Q}[G]$ -module M , we define

$$\mathrm{tor}_{\mathbb{Q}[G]}(M) = \bigcap \{\ker \phi : \phi : M \rightarrow \mathcal{U}(G) \text{ is a } \mathbb{Q}[G]\text{-homomorphism}\}.$$

We will show in Lemma 2.4 that, in the case where M is finitely generated, $\mathrm{tor}_{\mathbb{Q}[G]}(M)$ can equivalently be defined as

$$\mathrm{tor}_{\mathbb{Q}[G]}(M) = \bigcap \{\ker \phi : \phi : M \rightarrow \ell^2(G) \text{ is a } \mathbb{Q}[G]\text{-homomorphism}\}.$$

Moreover, if G is torsion-free and satisfies the strong Atiyah conjecture over $\mathbb{Q}$ (which states that the division closure $\mathcal{D}(G)$ of $\mathbb{Q}[G]$ in $\mathcal{U}(G)$ is a division ring), then

$$\mathrm{tor}_{\mathbb{Q}[G]}(M) = \bigcap \{\ker \phi : \phi : M \rightarrow \mathcal{D}(G) \text{ is a } \mathbb{Q}[G]\text{-homomorphism}\}.$$

Denote by ι_M the canonical map

$$\iota_M : M \longrightarrow \mathcal{D}(G) \otimes_{\mathbb{Q}[G]} M, \quad m \longmapsto 1 \otimes m.$$

Then any $\mathbb{Q}[G]$ -homomorphism $\phi : M \rightarrow \mathcal{D}(G)$ can be canonically decomposed as

$$\phi = \tilde{\phi} \circ \iota_M,$$

where $\tilde{\phi} : \mathcal{D}(G) \otimes_{\mathbb{Q}[G]} M \rightarrow \mathcal{D}(G)$ is a $\mathcal{D}(G)$ -homomorphism. Therefore,

$$\text{tor}_{\mathbb{Q}[G]}(M) = \ker(\iota_M).$$

We denote by $I_{\mathbb{Q}[G]}$ the augmentation ideal of $\mathbb{Q}[G]$ and if H is a subgroup of G we denote by ${}^G I_{\mathbb{Q}[H]}$ the left ideal of $\mathbb{Q}[G]$ generated by $I_{\mathbb{Q}[H]}$. Let L be a left $\mathbb{Q}[G]$ -submodule of $I_{\mathbb{Q}[G]}$ such that $L/{}^G I_{\mathbb{Q}[H]} = \text{tor}_{\mathbb{Q}[G]}(I_{\mathbb{Q}[G]}/{}^G I_{\mathbb{Q}[H]})$. A priori, the submodule L has nothing to do with the subgroup structure of G . If there exists a subgroup $\tilde{H}$ such that $L = {}^G I_{\mathbb{Q}[\tilde{H}]}$, then $\tilde{H}$ is unique and contains H (see Lemma 2.1), and we call this subgroup the L^2 -closure of H in G . We say that G is L^2 -subgroup rigid if for all finitely generated subgroups of G its L^2 -closure can be defined. In this paper we will show that free groups are L^2 -subgroup rigid.

Theorem 1.1. *Free groups are L^2 -subgroup rigid.*

It is very plausible that our proof can be extended to show that not only free groups, but also other groups are L^2 -subgroup rigid. We have chosen to explore this aspect in a forthcoming paper and, for simplicity, focus here only on the case of free groups. As we will see later, even in the case of free groups, we already can derive several important consequences.

A finitely generated subgroup H of a free group F is called **compressed** if for any finitely generated subgroup L of F containing H , $\text{rk}(H) \leq \text{rk}(L)$, and it is called **inert** if for any finitely generated subgroup L of F , $\text{rk}(H \cap L) \leq \text{rk}(L)$.

The concept of a subgroup being compressed is quite intuitive; it indicates that the subgroup is not contained in another subgroup of smaller rank. The concept of an inert subgroup was introduced in the work of Dicks and Ventura while they were studying equalizers in free groups [DV96].

We also say that H is L^2 -independent in F if the canonical map

$$\mathcal{D}(F) \otimes_{\mathbb{Q}[H]} I_{\mathbb{Q}[H]} \rightarrow \mathcal{D}(F) \otimes_{\mathbb{Q}[F]} I_{\mathbb{Q}[F]}$$

is injective. It is clear that inert subgroups are compressed and Antolin and Jaikin-Zapirain [AJZ22] proved that L^2 -independent subgroups are inert. On the other hand, Dicks and Ventura [DV96, Problem 1] (see also [KM18, Problem 19.102]) asked whether compressed subgroups are also inert and Antolin and Jaikin-Zapirain [AJZ22, Question 2] suggested that compressed subgroups may always be L^2 -independent. We confirm these two conjectures.

Corollary 1.2. *Let F be a free group and H a finitely generated subgroup. Then the following are equivalent:*

- (1) H is compressed in F ;
- (2) H is inert in F ;
- (3) H is L^2 -independent in F .

In Corollary 3.13 we will also show that these three properties of H are also equivalent of H being *strongly inert* in F (see Section 2.8 for the definition).

It is clear that the intersection of two inert subgroups is inert and by [MVW07, Theorem 3.9], it is algorithmically decidable, for a finitely generated subgroup of a free group, whether it is compressed. In [Iva18] Ivanov proved that it is algorithmically decidable whether a finitely generated subgroup of a free group is strongly

inert or not. The equivalence between inert and compressed subgroup provides us with the following immediate consequences.

Corollary 1.3. *The following holds.*

- (1) [KM18, Problem 19.103] *The intersection of two compressed subgroups of a free group is again compressed.*
- (2) [KM18, Problem 19.104] *It is algorithmically decidable, for a finitely generated subgroup of a free group, whether it is inert.*

A finitely generated subgroup H of a free group F is called **strictly compressed** in F if for any subgroup L of F , properly containing H , $\text{rk}(H) < \text{rk}(L)$ and **strictly inert** if for any subgroup L not contained in H , we have $\text{rk}(L \cap H) < \text{rk}(L)$. We say that H is L^2 -**closed** in G if it coincides with its own L^2 -closure. Another consequence of Theorem 1.1 is the following equivalence.

Corollary 1.4. *Let F a free group and H a finitely generated subgroup of F . Then the following are equivalent.*

- (1) *H is strictly compressed in F ;*
- (2) *H is strictly inert in F ;*
- (3) *H is L^2 -closed in F .*

We also obtain the following surprising property of the lattice of finitely generated subgroups of free groups.

Corollary 1.5. *Let F be a free group and H a finitely generated subgroup. Define*

$$\begin{aligned} \bar{\pi}(H \leq F) &= \min\{\text{rk}(L) : H \leq L \leq F\} \text{ and} \\ \text{Crit}(H \leq F) &= \{H \leq L \leq F : \text{rk}(L) = \bar{\pi}(H \leq F)\}. \end{aligned}$$

If L_1 and L_2 belong to $\text{Crit}(H \leq F)$, then $L_1 \cap L_2$ and $\langle L_1, L_2 \rangle$ also belong to $\text{Crit}(H \leq F)$.

We will see that the maximal subgroup in $\text{Crit}(H \leq F)$ is the L^2 -closure of H .

The paper is organized as follows. In Section 2, we describe preliminary results. The main step of our proof is found in Section 3.3, where we establish an analogue of Theorem 1.1 in characteristic $p = 2$. In this section, we explain why the prime $p = 2$ is particularly suitable for our argument. The case of characteristic 0 follows as a consequence of the case $p = 2$. In the last section, we propose several open questions for further research.

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2. PRELIMINARIES

2.1. General results. All of our rings R will be associative and unitary. All ring homomorphisms will map $1 \mapsto 1$. We will always assume that R has **invariant basis number** (this is, if the isomorphism of left R -modules $R^n \cong R^m$ implies $n = m$). The previous is ensured, for instance, when the ring R admits a ring

homomorphism to a field. We will write $\text{rk}(M) = n$ if M is a left R -module isomorphic to R^n .

By an **R -ring** we understand a ring homomorphism $\varphi : R \rightarrow S$. We will often refer to S as R -ring and omit the homomorphism φ if φ is clear from the context. Two R -rings $\varphi_1 : R \rightarrow S_1$ and $\varphi_2 : R \rightarrow S_2$ are said to be **isomorphic** if there exists a ring isomorphism $\alpha : S_1 \rightarrow S_2$ such that $\alpha \circ \varphi_1 = \varphi_2$.

If F is a finitely generated free group, we denote by $\text{rk}(F)$ the cardinality of free generators of F and define $\overline{\text{rk}}(F) = \max\{\text{rk}(F) - 1, 0\}$. In other words, $\overline{\text{rk}}(F)$ is typically $\text{rk}(F) - 1$, except when F is trivial, in which case $\overline{\text{rk}}(F) = 0$.

2.2. Left ideals associated with subgroups. In the introduction, we have defined the left ideal ${}^G I_{k[H]}$ of $k[G]$. It uniquely determines the subgroup H .

Lemma 2.1. *Let H and T be subgroups of a group G and k a commutative ring. If ${}^G I_{k[H]} \leq {}^G I_{k[T]}$, then $H \leq T$. In particular, if ${}^G I_{k[H]} = {}^G I_{k[T]}$, then $H = T$.*

Proof. For any $g \in G$, we put $\bar{g} = gH \in G/H$. It is clear that $\text{Stab}_G(\bar{1}) = H$. The left action of G on G/H induces a structure of left $k[G]$ -module on $k[G/H]$.

Since the trivial $k[H]$ -module k is isomorphic to $k[H]/I_{k[H]}$,

$$k[G]/{}^G I_{k[H]} \cong k[G] \otimes_{k[H]} k \cong k[G/H].$$

Moreover, the isomorphism is realized by the map $k[G]/{}^G I_{k[H]} \rightarrow k[G/H]$, which sends $1 + {}^G I_{k[H]}$ to $\bar{1}$. In particular,

$$\text{Stab}_G(1 + {}^G I_{k[H]}) = \text{Stab}_G(\bar{1}) = H.$$

Thus, if ${}^G I_{k[H]} \leq {}^G I_{k[T]}$, then $H = \text{Stab}_G(1 + {}^G I_{k[H]}) \leq \text{Stab}_G(1 + {}^G I_{k[T]}) = T$. $\square$

The following lemma gives an alternative description of the left $k[G]$ -module ${}^G I_{k[T]}/{}^G I_{k[H]}$.

Lemma 2.2. [JZ24, Lemma 2.1] *Let $H \leq T$ be subgroups of a group G and k a commutative ring. Then the canonical map*

$$k[G] \otimes_{k[T]} (I_{k[T]}/{}^T I_{k[H]}) \rightarrow {}^G I_{k[T]}/{}^G I_{k[H]},$$

sending $a \otimes (b + {}^T I_{k[H]})$ to $ab + {}^G I_{k[H]}$, is an isomorphism of left $k[G]$ -modules.

2.3. Algebraic subextensions. Let F be a free group and H a finitely generated subgroup of F . For a subgroup L of F containing H , we say that $H \leq L$ is an **algebraic extension** if there is no proper free factor of L containing H . We put

$$\mathcal{A}_{H \leq F} = \{L \leq F : H \leq L \text{ is an algebraic extension}\}.$$

The following result was proved by Takahasi [Tak51] (see also [MVW07]).

Proposition 2.3. *Let F be a free group and H a finitely generated subgroup. Then $\mathcal{A}_{H \leq F}$ is finite.*

2.4. The L^2 -torsion part of a left $\mathbb{Q}[G]$ -module. Let G be a countable group and let $\ell^2(G)$ denote the Hilbert space with Hilbert basis the elements of G , that is, $\ell^2(G)$ consists of all square-summable formal sums

$$\sum_{g \in G} a_g g$$

with $a_g \in \mathbb{C}$, and inner product

$$\left\langle \sum_{g \in G} a_g g, \sum_{g \in G} b_g g \right\rangle = \sum_{g \in G} a_g \overline{b_g}.$$

The left and right multiplication actions of G on itself extend to left and right actions of G on $\ell^2(G)$. The right action of G on $\ell^2(G)$ further extends to an action of elements of $\mathbb{Q}[G]$ on $\ell^2(G)$ as bounded linear operators on $\ell^2(G)$.

The von Neumann ring $\mathcal{N}(G)$ is the ring of bounded operators on $\ell^2(G)$ which commute with the left action of G . Thus, we can see $\mathbb{Q}[G]$ as a subalgebra of $\mathcal{N}(G)$. The ring $\mathcal{N}(G)$ satisfies the left and right Ore conditions (a result proved by S. K. Berberian in [Ber82]), and its classical ring of fractions is denoted by $\mathcal{U}(G)$. The ring $\mathcal{U}(G)$ can also be described as the ring of densely defined closed (unbounded) operators on $\ell^2(G)$ that commute with the left action of G (see [Lüc02, Kam19]).

As mentioned in the introduction, the Strong Atiyah Conjecture predicts that if G is torsion-free, then the division closure $\mathcal{D}(G)$ of $\mathbb{Q}[G]$ in $\mathcal{U}(G)$ is a division ring. In this paper we make use of Linnell's solution of the Strong Atiyah Conjecture for free groups [Lin93]. We also note that it was solved for all locally indicable groups in [JZLÁ19].

For a left $\mathbb{Q}[G]$ -module M , we define

$$\text{tor}_{\mathbb{Q}[G]}(M) = \bigcap \{\ker \phi : \phi : M \rightarrow \mathcal{U}(G) \text{ is a } \mathbb{Q}[G]\text{-homomorphism}\}.$$

Lemma 2.4. *Let G be a group and M a finitely generated left $\mathbb{Q}[G]$ -module.*

(1) *If M is finitely generated, then*

$$\text{tor}_{\mathbb{Q}[G]}(M) = \bigcap \{\ker \phi : \phi : M \rightarrow \ell^2(G) \text{ is a } \mathbb{Q}[G]\text{-homomorphism}\}.$$

(2) *if G is torsion-free and satisfies the strong Atiyah conjecture over $\mathbb{Q}$, then*

$$\text{tor}_{\mathbb{Q}[G]}(M) = \bigcap \{\ker \phi : \phi : M \rightarrow \mathcal{D}(G) \text{ is a } \mathbb{Q}[G]\text{-homomorphism}\}.$$

Proof. (1) Since $\mathcal{U}(G)$ is the classical Ore localization of $\mathcal{N}(G)$, for any finite set of elements $m_1, \dots, m_k \in \mathcal{U}(G)$ there exists an invertible element $u \in \mathcal{N}(G)$ such that

$$m_1 \cdot u, \dots, m_k \cdot u \in \mathcal{N}(G).$$

Since u is invertible, the left $\mathbb{Q}[G]$ -submodules of $\mathcal{U}(G)$ generated by $\{m_1, \dots, m_k\}$ and by $\{m_1 \cdot u, \dots, m_k \cdot u\}$ are isomorphic. Thus, any finitely generated submodule of $\mathcal{U}(G)$ is also a submodule of $\mathcal{N}(G)$.

Observe that the map

$$\mathcal{N}(G) \longrightarrow \ell^2(G), \quad \phi \mapsto \phi(1),$$

is an embedding (and moreover a $\mathbb{Q}[G]$ -homomorphism). On the other hand, for any $u \in \ell^2(G)$, we can define a densely defined closed (unbounded) operator on $\ell^2(G)$, ϕ_u that sends g to gu . This provides an embedding of $\ell^2(G)$ into $\mathcal{U}(G)$.

We conclude that the local structures of the left $\mathbb{Q}[G]$ -modules $\ell^2(G)$ and $\mathcal{U}(G)$ coincide, which implies the claim since M is finitely generated.

(2) Since $\mathcal{D}(G)$ is a $\mathbb{Q}[G]$ -submodule of $\mathcal{U}(G)$, it is clear that

$$\text{tor}_{\mathbb{Q}[G]}(M) \subseteq \bigcap \{\ker \phi : \phi : M \rightarrow \mathcal{D}(G) \text{ is a } \mathbb{Q}[G]\text{-homomorphism}\}.$$

To prove the converse inclusion, consider $m \in M$ and a $\mathbb{Q}[G]$ -homomorphism

$$\phi : M \longrightarrow \mathcal{U}(G)$$

such that $\phi(m) \neq 0$. Since $\mathcal{D}(G)$ is a division ring and $\mathcal{U}(G)$ is a $\mathcal{D}(G)$ -space, there exists a $\mathcal{D}(G)$ -homomorphism

$$\psi : \mathcal{U}(G) \longrightarrow \mathcal{D}(G)$$

with $\psi(\phi(m)) \neq 0$. Therefore,

$$m \notin \bigcap \{ \ker \phi : \phi : M \rightarrow \mathcal{D}(G) \text{ is a } \mathbb{Q}[G]\text{-homomorphism} \}. \quad \square$$

2.5. Universal division ring of fractions of group rings. Let k be a commutative domain and G a group. A **division $k[G]$ -ring of fractions** is an embedding $k[G] \hookrightarrow \mathcal{D}$ of $k[G]$ into a division ring $\mathcal{D}$ such that the elements of $k[G]$ generate $\mathcal{D}$ as a division ring. If G satisfies the Strong Atiyah Conjecture, then the embedding

$$\mathbb{Q}[G] \hookrightarrow \mathcal{D}(G)$$

is an example of a division $\mathbb{Q}[G]$ -ring of fractions.

In this paper we will work with a division $k[G]$ -ring of fractions $\mathcal{D}$ satisfying the following additional properties:

- (P1) For every division $k[G]$ -ring $\mathcal{E}$, and for any left finitely presented $k[G]$ -module M , $\dim_{\mathcal{D}} \mathcal{D} \otimes_{k[G]} M \leq \dim_{\mathcal{E}} \mathcal{E} \otimes_{k[G]} M$. In this case we say that the embedding $k[G] \hookrightarrow \mathcal{D}$ is the **universal division $k[G]$ -ring of fractions**. By definition, it is uniquely determined up to $k[G]$ -isomorphism (see [Mal80]); to emphasize this, we denote $\mathcal{D}$ by $\mathcal{D}_{k[G]}$. If Q is the field of fractions of k , then $\mathcal{D}_{k[G]}$ is automatically universal for $Q[G]$; in other words, $\mathcal{D}_{Q[G]} \cong \mathcal{D}_{k[G]}$ as $Q[G]$ -rings. We define

$$\beta_i^{k[G]}(M) = \dim_{\mathcal{D}_{k[G]}} \mathrm{Tor}_i(\mathcal{D}_{k[G]}, M).$$

- (P2) $\mathcal{D}_{Q[G]}$ is of weak dimension 1 as a right $Q[G]$ -module. This means that for any left $Q[G]$ -module M , $\beta_i^{Q[G]}(M) = 0$ for $i > 1$.
- (P3) For any left $Q[G]$ -submodule M of $\mathcal{D}_{Q[G]}$, $\beta_1^{Q[G]}(M) = 0$.
- (P4) The division $k[G]$ -ring $\mathcal{D}_{k[G]}$ is **Lewin**, i.e. for any subgroup H of G , if we denote by $\mathcal{D}_H$ the division closure of $k[H]$ in $\mathcal{D}_{k[G]}$, then the map

$$\mathcal{D}_H \otimes_{k[H]} k[G] \rightarrow \mathcal{D}_{k[G]}, \quad d \otimes a \mapsto da \quad (d \in \mathcal{D}_H, a \in k[H]),$$

is injective.

If k is a field or $\mathbb{Z}$ and F is a free group, then the group ring $k[F]$ has an embedding in a division ring satisfying the properties (P1)-(P3), because it is a Sylvester domain (see [DS78]). Moreover, it also satisfies (P4) (see, for example, [JZ21, Subsection 2.3]).

We will need the following consequence of Property (P1) for $\mathbb{Z}[F]$. Let M be a left finitely presented $\mathbb{Z}[F]$ -module. Since the embedding $\mathbb{Z}[F] \hookrightarrow \mathcal{D}_{\mathbb{Z}[F]}$ is universal and $\mathcal{D}_{\mathbb{Z}[F]} = \mathcal{D}_{\mathbb{Q}[F]}$,

$$\dim_{\mathcal{D}_{\mathbb{Z}[F]}} \mathcal{D}_{\mathbb{Z}[F]} \otimes_{\mathbb{Z}[F]} M \leq \dim_{\mathcal{D}_{\mathbb{F}_p[F]}} \mathcal{D}_{\mathbb{F}_p[F]} \otimes_{\mathbb{Z}[F]} M$$

and we conclude that

$$\begin{aligned}
 (1) \quad \beta_0^{\mathbb{Q}[F]}(\mathbb{Q} \otimes_{\mathbb{Z}} M) &= \dim_{\mathcal{D}_{\mathbb{Q}[F]}} \mathcal{D}_{\mathbb{Q}[F]} \otimes_{\mathbb{Q}[F]} (\mathbb{Q} \otimes_{\mathbb{Z}} M) = \\
 &\dim_{\mathcal{D}_{\mathbb{Z}[F]}} \mathcal{D}_{\mathbb{Z}[F]} \otimes_{\mathbb{Z}[F]} M \leq \dim_{\mathcal{D}_{\mathbb{F}_p[F]}} \mathcal{D}_{\mathbb{F}_p[F]} \otimes_{\mathbb{Z}[F]} M = \\
 &\dim_{\mathcal{D}_{\mathbb{F}_p[F]}} \mathcal{D}_{\mathbb{F}_p[F]} \otimes_{\mathbb{F}_p[F]} (\mathbb{F}_p \otimes_{\mathbb{Z}} M) = \beta_0^{\mathbb{F}_p[F]}(\mathbb{F}_p \otimes_{\mathbb{Z}} M).
 \end{aligned}$$

Given an extension of fields $K_1 \leq K_2$, we also have that the division closure of $K_1[F]$ in $\mathcal{D}_{K_2[F]}$ is isomorphic (as a $K_1[F]$ -ring) to $\mathcal{D}_{K_1[F]}$. Thus, we obtain the following immediate consequence.

Proposition 2.5. *Let $K_1 \leq K_2$ be an extension of fields and M a left $K_1[F]$ -module. Then for any $i \geq 0$, $\beta_i^{K_2[F]}(K_2 \otimes_{K_1} M) = \beta_i^{K_1[F]}(M)$.*

Let K be a field. The property (P4) has its origin in a work of Hughes [Hug70]. The result of Hughes implies that if G is locally indicable, then there exists at most one division $K[G]$ -ring of fractions satisfying (P4). In the case of free group F this has two consequences. First, for any subgroup H of F , $\mathcal{D}_H$ is isomorphic to $\mathcal{D}_{K[H]}$ as a $K[H]$ -ring and second that the division $\mathbb{Q}[F]$ -ring $\mathcal{D}(F)$ mentioned in the introduction is isomorphic to $\mathcal{D}_{\mathbb{Q}[F]}$ as a $\mathbb{Q}[F]$ -ring. Thus, for any left $\mathbb{Q}[F]$ -module M we can define its *i th L^2 -Betti numbers* in a pure algebraic way as $\beta_i^{\mathbb{Q}[F]}(M)$. By analogy, if M is a left $\mathbb{F}_p[F]$ -module $\beta_i^{\mathbb{F}_p[F]}(M)$ are called the *i th mod- p L^2 -Betti numbers* of M .

2.6. The Euler characteristic. A R -module M is of type FL (resp. FP) if it has a finite resolution consisting of finitely generated free (resp. projective) R -modules.

For a R -module M of type FL that has the following resolution

$$0 \longrightarrow R^{n_k} \longrightarrow \cdots \longrightarrow R^{n_0} \longrightarrow M \longrightarrow 0,$$

we define the Euler characteristic of M as

$$\chi^R(M) = \sum_{i=0}^k (-1)^i n_i.$$

A standard application of Schanuel's lemma [Bro82, Chapter VIII, Lemma 4.4] implies, that this definition does not depend on the free resolution, and, moreover, if $0 \rightarrow M_1 \rightarrow M_2 \rightarrow M_3 \rightarrow 0$ is an exact sequence of FL R -modules, then

$$(2) \quad \chi^R(M_2) = \chi^R(M_1) + \chi^R(M_3).$$

If $R \rightarrow \mathcal{D}$ is a division R -ring, the $\mathcal{D}$ -Betti numbers of a left R -module M are defined as

$$b_i^{R, \mathcal{D}}(M) = \dim_{\mathcal{D}} \operatorname{Tor}_i^R(\mathcal{D}, M).$$

If M is of type FL, then using the long exact sequence of Tor-functors one obtains that the Euler characteristic of M can be calculated as

$$(3) \quad \chi^R(M) = \sum_i (-1)^i b_i^{R, \mathcal{D}}(M).$$

Proposition 2.6. [Bie81, Proposition 1.4 and Proposition 4.1b] *Let $0 \rightarrow M_1 \rightarrow M_2 \rightarrow M_3 \rightarrow 0$ be an exact sequence of R -modules. If two of the modules M_1 , M_2 or M_3 are of type FP, then the third one is also of type FP.*

The same statement can also be obtained for **FL**-modules using [Bou07, Theorem 3.9.1]. However, we will not use this fact, since all modules of type **FP** considered in this paper are also of type **FL**.

In the case of rings $k[G]$ satisfying the properties (P1)-(P3) we have the following result.

Proposition 2.7. *Assume $R = k[G]$ be a group ring having an embedding in a division ring satisfying the properties (P1)-(P3) and let M be a left **FL** R -module. Then*

$$\chi^R(M) = \beta_0^R(M) - \beta_1^R(M).$$

*In particular, for any non-trivial left **FL** R -submodule M of $\mathcal{D}_R$, $\chi^R(M) > 0$.*

Proof. From Eq. (3) we obtain that

$$\chi^R(M) = \sum_{i=0}^k (-1)^i \beta_i^R(M).$$

It then follows from Property (P2) that

$$\chi^R(M) = \beta_0^R(M) - \beta_1^R(M),$$

and from Property (P3) that if M is a left R -submodule of $\mathcal{D}_R$, then

$$\chi^R(M) = \beta_0^R(M).$$

If M is a non-trivial submodule of $\mathcal{D}_R$, we have $\beta_0^R(M) = \dim_{\mathcal{D}_R} \mathcal{D}_R \otimes_R M \geq 1$. $\square$

2.7. Modules over group algebra over a free group.

Proposition 2.8. *Let K be a field and F a free group. Then the ring $K[F]$ has global dimension 1, and every finitely presented $K[F]$ -module is of type **FL**.*

Moreover, if M is a finitely generated $K[F]$ -module and $\beta_1^{K[F]}(M) < \infty$, then there exists an exact sequence

$$0 \rightarrow K[F]^r \rightarrow K[F]^d \rightarrow M \rightarrow 0.$$

Proof. The first part follows from [Coh64, Corollary 5], where it is shown that every submodule of a free $K[F]$ -module is free.

For the second part of the proposition, we have an exact sequence

$$0 \longrightarrow L \longrightarrow K[F]^d \longrightarrow M \longrightarrow 0,$$

where L is a free $K[F]$ -module. The condition $\beta_1^{K[F]}(M) < \infty$ implies that $\beta_0^{K[F]}(L) < \infty$, and so, L is finitely generated. $\square$

As a corollary of the previous proposition, we obtain the following result.

Lemma 2.9. *Let K be a field, F a free group, $M_1 \leq M_2$ two left $K[F]$ -modules and N a right $K[F]$ -module. Then the natural map*

$$\mathrm{Tor}_1^{K[F]}(N, M_1) \rightarrow \mathrm{Tor}_1^{K[F]}(N, M_2)$$

is injective.

Proof. Using the long exact sequence for Tor functors, we obtain that the kernel of this map is $\mathrm{Tor}_2^{K[F]}(N, M_2/M_1)$, which is zero since $K[F]$ has global dimension 1. $\square$

2.8. $\mathcal{D}_{K[F]}$ -independent modules. Let F be a free group and K a field. By analogy with L^2 -independence, we also say that a subgroup H is $\mathcal{D}_{K[F]}$ -independent in F if the canonical map

$$\mathcal{D}_{K[F]} \otimes_{K[H]} I_{K[H]} \rightarrow \mathcal{D}_{K[F]} \otimes_{K[F]} I_{K[F]}$$

is injective. This is equivalent to the condition $\beta_1^{K[F]}(I_{K[F]}/^F I_{K[H]}) = 0$. Observe that, since $\mathcal{D}(F)$ is isomorphic as a $\mathbb{Q}[F]$ -ring to $\mathcal{D}_{\mathbb{Q}[F]}$, $\mathcal{D}_{\mathbb{Q}[F]}$ -independence is the same as L^2 -independence defined before.

If H is a finitely generated subgroup of a free group F , we say that H is **strongly inert** in F if for any finitely generated subgroup U of F we have that

$$\sum_{x \in U \setminus F/H} \overline{\text{rk}}(U \cap xHx^{-1}) \leq \overline{\text{rk}}(U).$$

This notion has been introduced by Ivanov [Iva18]. It is clear that a strongly inert subgroup of a free group is also inert. The proof of the following proposition is a small modification of the proof of [AJZ22, Proposition 5.2].

Proposition 2.10. *Let H be a finitely generated subgroup of a finitely generated free group F . Assume that H is $\mathcal{D}_{K[F]}$ -independent in F . Then H is strongly inert in F .*

Proof. Let U be a finitely generated subgroup of F . Since H is $\mathcal{D}_{K[F]}$ -independent in F

$$\text{Tor}_1^{K[F]}(\mathcal{D}_{K[F]}, I_{K[F]}/^F I_{K[H]}) = 0.$$

By Property (P4), the right $K[F]$ -module $\mathcal{D}_{K[U]} \otimes_{K[U]} K[F]$ naturally embeds into $\mathcal{D}_{K[F]}$. Hence, by Lemma 2.9, we also have

$$\text{Tor}_1^{K[F]}(\mathcal{D}_{K[U]} \otimes_{K[U]} K[F], I_{K[F]}/^F I_{K[H]}) = 0.$$

Thus, by Shapiro's Lemma

$$(4) \quad \begin{aligned} \text{Tor}_1^{K[U]}(\mathcal{D}_{K[U]}, I_{K[F]}/^F I_{K[H]}) &= \\ \text{Tor}_1^{K[F]}(\mathcal{D}_{K[U]} \otimes_{K[U]} K[F], I_{K[F]}/^F I_{K[H]}) &= 0. \end{aligned}$$

As we have shown in the proof of Lemma 2.1, the left $K[F]$ -modules $K[F]/^F I_{K[H]}$ and $K[F/H]$ are isomorphic. Thus, we have the following isomorphism of left $K[U]$ -modules.

$$K[F]/^F I_{K[H]} \cong K[F/H] \cong \bigoplus_{x \in U \setminus F/H} K[U/(U \cap xHx^{-1})].$$

Since, $\overline{\text{rk}}(U \cap xHx^{-1}) = \dim_{\mathcal{D}_{K[U]}} \text{Tor}_1^{K[U]}(\mathcal{D}_{K[U]}, K[U/(U \cap xHx^{-1})])$, we obtain that

$$\sum_{x \in U \setminus F/H} \overline{\text{rk}}(U \cap xHx^{-1}) = \dim_{\mathcal{D}_{K[U]}} \text{Tor}_1^{K[U]}(\mathcal{D}_{K[U]}, K[F]/^F I_{K[H]}).$$

From the exact sequence of the left $K[U]$ -modules

$$0 \rightarrow I_{K[F]}/^F I_{K[H]} \rightarrow K[F]/^F I_{K[H]} \rightarrow K \rightarrow 0,$$

we obtain that

$$\begin{aligned} \dim_{\mathcal{D}_{K[U]}} \mathrm{Tor}_1^{K[U]}(\mathcal{D}_{K[U]}, K[F]/^F I_{K[H]}) &\leq \\ \dim_{\mathcal{D}_{K[U]}} \mathrm{Tor}_1^{K[U]}(\mathcal{D}_{K[U]}, I_{K[F]}/^F I_{K[H]}) + \dim_{\mathcal{D}_{K[U]}} \mathrm{Tor}_1^{K[U]}(\mathcal{D}_{K[U]}, K) &\stackrel{\text{Eq. (4)}}{=} \\ \dim_{\mathcal{D}_{K[U]}} \mathrm{Tor}_1^{K[U]}(\mathcal{D}_{K[U]}, K) &= \overline{\mathrm{rk}}(U). \end{aligned}$$

This gives the desired inequality:

$$\sum_{x \in U \setminus F/H} \overline{\mathrm{rk}}(U \cap xHx^{-1}) \leq \overline{\mathrm{rk}}(U). \quad \square$$

3. PROOF OF THE RESULTS

In this section we assume that K is a field and F is a free group.

3.1. The $\mathcal{D}_{K[F]}$ -torsion part of a left $K[F]$ -module. Let k be either a field K or $k = \mathbb{Z}$, and let M be a left $k[F]$ -module. Denote by $\mathrm{tor}_{k[F]}(M)$ the kernel of the canonical map

$$M \longrightarrow \mathcal{D}_{k[F]} \otimes_{k[F]} M.$$

The next result shows that $\mathrm{tor}_{K[F]}(M)$ is of type FL if M is of type FL.

Proposition 3.1. *Let M be a left finitely presented $K[F]$ -module. Then $\mathrm{tor}_{K[F]}(M)$ is of type FL.*

Proof. By Property (P3), $\beta_1^{K[F]}(M/\mathrm{tor}_{K[F]}(M)) = 0$. Hence, by Proposition 2.8, M and $M/\mathrm{tor}_{K[F]}(M)$ are of type FL. Hence, by Proposition 2.6, $\mathrm{tor}_{K[F]}(M)$ is of type FP and so it is of type FL, by Proposition 2.8. $\square$

The following result gives a characterization of $\mathrm{tor}_{K[F]}(M)$.

Proposition 3.2. *Let M be a left finitely presented $K[F]$ -module. Then we have that*

$$\beta_0^{K[F]}(\mathrm{tor}_{K[F]}(M)) = 0.$$

Moreover, $\mathrm{tor}_{K[F]}(M)$ is the maximal submodule of M with this property and we also have that

$$\beta_1^{K[F]}(\mathrm{tor}_{K[F]}(M)) = \beta_1^{K[F]}(M),$$

and so, $\chi^{K[F]}(\mathrm{tor}_{K[F]}(M)) = -\beta_1^{K[F]}(M)$.

Proof. Let $M_1 = \mathrm{tor}_{K[F]}(M)$. By Proposition 3.1, it is a left $K[F]$ -module of type FL. Observe that

$$\beta_0^{K[F]}(M) = \beta_0^{K[F]}(M/M_1),$$

and by property (P3), we also have $\beta_1^{K[F]}(M/M_1) = 0$. The short exact sequence

$$0 \longrightarrow M_1 \longrightarrow M \longrightarrow M/M_1 \longrightarrow 0$$

induces the long exact sequence in Tor:

$$\begin{aligned} 0 \rightarrow \mathrm{Tor}_1^{\mathcal{D}_{K[F]}}(\mathcal{D}_{K[F]}, M_1) \rightarrow \mathrm{Tor}_1^{\mathcal{D}_{K[F]}}(\mathcal{D}_{K[F]}, M) \rightarrow \mathrm{Tor}_1^{\mathcal{D}_{K[F]}}(\mathcal{D}_{K[F]}, M/M_1) \rightarrow \\ \mathrm{Tor}_0^{\mathcal{D}_{K[F]}}(\mathcal{D}_{K[F]}, M_1) \rightarrow \mathrm{Tor}_0^{\mathcal{D}_{K[F]}}(\mathcal{D}_{K[F]}, M) \rightarrow \mathrm{Tor}_0^{\mathcal{D}_{K[F]}}(\mathcal{D}_{K[F]}, M/M_1) \rightarrow 0. \end{aligned}$$

Hence, we obtain

$$\beta_0^{K[F]}(M_1) = 0 \quad \text{and} \quad \beta_1^{K[F]}(M_1) = \beta_1^{K[F]}(M).$$

Finally, it is clear that if $\beta_0^{K[F]}(N) = 0$, then $N \leq M_1$. $\square$

3.2. $\mathcal{D}_{K[F]}$ -torsion-free modules. We say that M is $\mathcal{D}_{K[F]}$ -torsion-free if

$$\text{tor}_{K[F]}(M) = 0.$$

The following result is similar to [JZ24, Lemma 4.7].

Proposition 3.3. *Let K be a field, H a subgroup of a free group F and M a $\mathcal{D}_{K[H]}$ -torsion-free left $K[H]$ -module. Then $K[F] \otimes_{K[H]} M$ is $\mathcal{D}_{K[F]}$ -torsion-free.*

Proof. Let $\mathcal{D}_H$ be the division closure of $K[H]$ in $\mathcal{D}_{K[F]}$. As we mentioned in Section 2.5, $\mathcal{D}_H$ and $\mathcal{D}_{K[H]}$ are isomorphic as $K[H]$ -rings. Therefore, the map $M \rightarrow \mathcal{D}_H \otimes_{K[H]} M$ is injective. Then, since $K[F]$ is a free right $K[H]$ -module, the map

$$K[F] \otimes_{K[H]} M \xrightarrow{\alpha} K[F] \otimes_{K[H]} (\mathcal{D}_H \otimes_{K[H]} M)$$

is also injective.

Consider the canonical isomorphism between tensor products

$$K[F] \otimes_{K[H]} (\mathcal{D}_H \otimes_{K[H]} M) \xrightarrow{\beta} (K[F] \otimes_{K[H]} \mathcal{D}_H) \otimes_{K[H]} M.$$

By Property (P4), the canonical map

$$K[F] \otimes_{K[H]} \mathcal{D}_H \rightarrow \mathcal{D}_{K[F]}$$

is injective. Moreover, since $\mathcal{D}_H$ is a division ring, the image of $K[F] \otimes_{K[H]} \mathcal{D}_H$ is a direct summand of $\mathcal{D}_{K[F]}$ as a right $\mathcal{D}_H$ -submodule (and so, it is also a direct summand as a right $K[H]$ -submodule). Thus, the following canonical map of $K[F]$ -modules

$$(K[F] \otimes_{K[H]} \mathcal{D}_H) \otimes_{K[H]} M \xrightarrow{\gamma} \mathcal{D}_{K[F]} \otimes_{K[H]} M$$

is injective. Hence $\varphi = \gamma \circ \beta \circ \alpha$ is an injective homomorphism of left $K[F]$ -modules. Since a submodule of a $\mathcal{D}_{K[F]}$ -torsion-free module is also $\mathcal{D}_{K[F]}$ -torsion-free, $K[F] \otimes_{K[H]} M$ is $\mathcal{D}_{K[F]}$ -torsion-free. $\square$

Observe that if K_1 is a subfield of a field K_2 , then the division closure of $K_1[F]$ in $\mathcal{D}_{K_2[F]}$ is isomorphic to $\mathcal{D}_{K_1[F]}$ as a $K_1[F]$ -ring. This implies the following consequence.

Lemma 3.4. *Let K_1 be a subfield of a field K_2 and let F be a free group. If a $K_1[F]$ -module M is $\mathcal{D}_{K_1[F]}$ -torsion-free, then the $K_2[F]$ -module $K_2 \otimes_{K_1} M$ is $\mathcal{D}_{K_2[F]}$ -torsion-free.*

3.3. $\mathcal{D}_{\mathbb{F}_2[F]}$ -subgroup rigidity of free groups. We say that F is $\mathcal{D}_{K[F]}$ -subgroup rigid if for any finitely generated subgroup H of F , there exists a subgroup $\tilde{H}$ of F containing H such that

$$\text{tor}_{K[F]}(I_{K[F]}/^F I_{K[H]}) = {}^F I_{K[\tilde{H}]} / {}^F I_{K[H]}.$$

In this section we prove the following analog of Theorem 1.1.

Theorem 3.5. *Let F be a free group. Then F is $\mathcal{D}_{\mathbb{F}_2[F]}$ -subgroup rigid.*

First we will prove an auxiliary result. It describes the $K[F]$ -submodules of $I_{K[F]}$ of codimension 1 in the case $K = \mathbb{F}_2$. This is where we strongly use the condition $K = \mathbb{F}_2$.

Lemma 3.6. *Let F be a finitely generated free group. Let N be a left $\mathbb{F}_2[F]$ -submodule of $I_{\mathbb{F}_2[F]}$ such that $I_{\mathbb{F}_2[F]}/N \cong \mathbb{F}_2$ is the trivial $\mathbb{F}_2[F]$ -module of dimension 1 over $\mathbb{F}_2$. Then there exists a subgroup U of F of index 2 such that $N = {}^F I_{\mathbb{F}_2[U]}$.*

Proof. Consider the set $\mathcal{S}$ of left $\mathbb{F}_2[F]$ -submodules N of $I_{\mathbb{F}_2[F]}$ such that $I_{\mathbb{F}_2[F]}/N \cong \mathbb{F}_2$. Then $\mathcal{S}$ consists of the kernels of non-trivial $\mathbb{F}_2[F]$ -homomorphisms from $I_{\mathbb{F}_2[F]}$ to $\mathbb{F}_2$. Since, $\text{rk}(I_{\mathbb{F}_2[F]}) = \text{rk}(F)$, $\mathcal{S}$ has $2^{\text{rk}(F)} - 1$ submodules. If U is a subgroup of F of index 2, $\dim_{\mathbb{F}_2} \mathbb{F}_2[F]/{}^F I_{\mathbb{F}_2[U]} = 2$, and so, $\dim_{\mathbb{F}_2} I_{\mathbb{F}_2[F]}/{}^F I_{\mathbb{F}_2[U]} = 1$. Thus, ${}^F I_{\mathbb{F}_2[U]} \in \mathcal{S}$. Since, the number of subgroups of F of index 2 is $2^{\text{rk}(F)} - 1$, we obtain the lemma. $\square$

Proof of Theorem 3.5. We start the proof with the following useful claim.

Claim 3.7. *Let $H \leq U \leq L \leq F$. Then the $\mathbb{F}_2[L]$ -module*

$$\text{tor}_{\mathbb{F}_2[L]}({}^L I_{\mathbb{F}_2[U]} / {}^L I_{\mathbb{F}_2[H]})$$

is equal to the $\mathbb{F}_2[L]$ -submodule of ${}^L I_{\mathbb{F}_2[U]} / {}^L I_{\mathbb{F}_2[H]}$ generated by

$$\text{tor}_{\mathbb{F}_2[U]}(I_{\mathbb{F}_2[U]} / {}^U I_{\mathbb{F}_2[H]}).$$

Proof. Recall that $\mathcal{D}_U$ denotes the division closure of $\mathbb{F}_2[U]$ in $\mathcal{D}_{\mathbb{F}_2[F]}$. Since $\mathcal{D}_U$ is isomorphic to $\mathcal{D}_{\mathbb{F}_2[U]}$ as a $\mathbb{F}_2[U]$ -ring, we have that

$$\text{tor}_{\mathbb{F}_2[U]}(I_{\mathbb{F}_2[U]} / {}^U I_{\mathbb{F}_2[H]}) \leq \text{tor}_{\mathbb{F}_2[L]}({}^L I_{\mathbb{F}_2[U]} / {}^L I_{\mathbb{F}_2[H]}).$$

On the other hand, let $N_0 = I_{\mathbb{F}_2[U]} / {}^U I_{\mathbb{F}_2[H]}$, $N = {}^L I_{\mathbb{F}_2[U]} / {}^L I_{\mathbb{F}_2[H]}$ and N_1 be the $\mathbb{F}_2[L]$ -submodule of N generated by $\text{tor}_{\mathbb{F}_2[U]}(N_0)$. By Lemma 2.2, we know that $N \cong \mathbb{F}_2[L] \otimes_{\mathbb{F}_2[U]} N_0$. Hence N/N_1 is isomorphic to $\mathbb{F}_2[L] \otimes_{\mathbb{F}_2[U]} (N_0 / \text{tor}_{\mathbb{F}_2[U]}(N_0))$. By Proposition 3.3, N/N_1 is $\mathcal{D}_{\mathbb{F}_2[U]}$ -torsion-free. Hence $\text{tor}_{\mathbb{F}_2[L]}(N) \leq N_1$. $\diamond$

Let H be a finitely generated subgroup of F . Define

$$\mathcal{S} = \{H \leq L \leq F : \text{there exists no subgroup } \tilde{H} \leq L \text{ containing } H \text{ such that}$$

$$\text{tor}_{\mathbb{F}_2[L]}(I_{\mathbb{F}_2[L]} / {}^L I_{\mathbb{F}_2[H]}) \text{ is equal to } {}^L I_{\mathbb{F}_2[\tilde{H}]} / {}^L I_{\mathbb{F}_2[H]}\}.$$

Claim 3.8. *Let $H \leq L \leq F$ and let U be a free factor of L containing H . If $L \in \mathcal{S}$, then $U \in \mathcal{S}$. In particular, if $\mathcal{S} \neq \emptyset$, then $\mathcal{S} \cap \mathcal{A}_{H \leq L} \neq \emptyset$.*

Proof. Let $U \leq L$ such that $L = U * U_1$. Then

$$I_{\mathbb{F}_2[L]} / {}^L I_{\mathbb{F}_2[H]} = ({}^L I_{\mathbb{F}_2[U]} / {}^L I_{\mathbb{F}_2[H]}) \oplus {}^L I_{\mathbb{F}_2[U_1]}.$$

Hence $\text{tor}_{\mathbb{F}_2[L]}(I_{\mathbb{F}_2[L]} / {}^L I_{\mathbb{F}_2[H]}) = \text{tor}_{\mathbb{F}_2[F]}({}^L I_{\mathbb{F}_2[U]} / {}^L I_{\mathbb{F}_2[H]}),$ and so by Claim 3.7, $U \in \mathcal{S}$. $\diamond$

Assume that $F \in \mathcal{S}$. We want to obtain a contradiction. By Claim 3.8, $\mathcal{S} \cap \mathcal{A}_{H \leq F} \neq \emptyset$. By Proposition 2.3 there exists a minimal subgroup L in $\mathcal{S} \cap \mathcal{A}_{H \leq F}$. In particular L is finitely generated.

Put $M = I_{\mathbb{F}_2[L]} / {}^L I_{\mathbb{F}_2[H]}$. Since $L \in \mathcal{S}$, $\text{tor}_{\mathbb{F}_2[L]}(M) \neq \{0\}$.

Claim 3.9. *There exists a subgroup U of L of index 2 such that*

$$\text{tor}_{\mathbb{F}_2[L]}(M) \leq {}^L I_{\mathbb{F}_2[U]} / {}^L I_{\mathbb{F}_2[H]}.$$

Proof. Since $L \in \mathcal{S}$, $\text{tor}_{\mathbb{F}_2[L]}(M) \neq M$. Therefore,

$$\beta_0^{\mathbb{F}_2[L]}(M/\text{tor}_{\mathbb{F}_2[L]}(M)) > 0.$$

Therefore, by Property (P1), $\mathbb{F}_2 \otimes_{\mathbb{F}_2[L]} (M/\text{tor}_{\mathbb{F}_2[L]}(M))$ is not trivial. Hence there exists a left $\mathbb{F}_2[L]$ -submodule N of M such that $\text{tor}_{\mathbb{F}_2[L]}(M) \leq N < M$ and $M/N \cong \mathbb{F}_2$. By Lemma 3.6, $N = {}^L I_{\mathbb{F}_2[U]}/{}^L I_{\mathbb{F}_2[H]}$ for some subgroup U of index 2 in L . $\diamond$

Take U from the previous claim. From Proposition 3.2 we obtain that

$$(5) \quad \text{tor}_{\mathbb{F}_2[L]}(M) = \text{tor}_{\mathbb{F}_2[L]}({}^L I_{\mathbb{F}_2[U]}/{}^L I_{\mathbb{F}_2[H]}).$$

By our choice of L and Claim 3.8, there exists a subgroup $\tilde{H}$ of U containing H , such that

$$(6) \quad \text{tor}_{\mathbb{F}_2[U]}(I_{\mathbb{F}_2[U]}/{}^U I_{\mathbb{F}_2[H]}) = {}^U I_{\mathbb{F}_2[\tilde{H}]} / {}^U I_{\mathbb{F}_2[H]}.$$

Hence,

$$\text{tor}_{\mathbb{F}_2[L]}(M) \stackrel{\text{Eq. (5)}}{=} \text{tor}_{\mathbb{F}_2[L]}({}^L I_{\mathbb{F}_2[U]}/{}^L I_{\mathbb{F}_2[H]}) \stackrel{\text{Claim 3.7 and Eq. (6)}}{=} {}^L I_{\mathbb{F}_2[\tilde{H}]} / {}^L I_{\mathbb{F}_2[H]}.$$

This is a contradiction that finishes the proof of the theorem. $\square$

Let K be a field and H a subgroup of a free group F . Assume that there exists a subgroup $\tilde{H}$ of F containing H such that

$$\text{tor}_{K[F]}(I_{K[F]}/{}^F I_{K[H]}) = {}^F I_{K[\tilde{H}]} / {}^F I_{K[H]}.$$

By Lemma 2.1, $\tilde{H}$ is unique and we say that $\tilde{H}$ is the $\mathcal{D}_{K[F]}$ -closure of H . By Lemma 2.4, the notion of L^2 -closure introduced in the introduction and the notion of $\mathcal{D}_{\mathbb{Q}[F]}$ -closure coincide.

Theorem 3.5 implies that for every finitely generated subgroup H of G the $\mathcal{D}_{\mathbb{F}_2[F]}$ -closure of H can be defined. In fact, we can also conclude that the $\mathcal{D}_{\mathbb{F}_2[F]}$ -closure of an arbitrary subgroup H of F can be defined. Indeed, let $H = \bigcup H_i$ be expressed as a union of finitely generated subgroups. Then the $\mathcal{D}_{\mathbb{F}_2[F]}$ -closure of H coincides with the union of the $\mathcal{D}_{\mathbb{F}_2[F]}$ -closures of the subgroups H_i .

By Lemma 3.4, if K is a field of characteristic 2, then $\mathcal{D}_{K[F]}$ -closure of H coincides with its $\mathcal{D}_{\mathbb{F}_2[F]}$ -closure.

3.4. Proof of Theorem 1.1. Let F be a free group and H a finitely generated subgroup of F . Let $\tilde{H}$ be the $\mathcal{D}_{\mathbb{F}_2[F]}$ -closure of H . In particular by Proposition 3.2, we have that

$$(7) \quad \beta_0^{\mathbb{F}_2[F]}({}^F I_{\mathbb{F}_2[\tilde{H}]} / {}^F I_{\mathbb{F}_2[H]}) = 0.$$

We want to show that

$$\text{tor}_{\mathbb{Q}[F]}(I_{\mathbb{Q}[F]}/{}^F I_{\mathbb{Q}[H]}) = {}^F I_{\mathbb{Q}[\tilde{H}]} / {}^F I_{\mathbb{Q}[H]}.$$

This will follow from the following two claims.

Claim 3.10. *We have that $\beta_0^{\mathbb{Q}[F]}({}^F I_{\mathbb{Q}[\tilde{H}]} / {}^F I_{\mathbb{Q}[H]}) = 0$. In particular,*

$${}^F I_{\mathbb{Q}[\tilde{H}]} / {}^F I_{\mathbb{Q}[H]} \leq \text{tor}_{\mathbb{Q}[F]}(I_{\mathbb{Q}[F]}/{}^F I_{\mathbb{Q}[H]}).$$

Proof. Consider the left $\mathbb{Z}[F]$ -module $N = {}^F I_{\mathbb{Z}[\tilde{H}]} / {}^F I_{\mathbb{Z}[H]}$. Then we obtain that

$$\beta_0^{\mathbb{Q}[F]}(\mathbb{Q} \otimes_{\mathbb{Z}} N) \stackrel{\text{Eq. (1)}}{\leq} \beta_0^{\mathbb{F}_2[F]}(\mathbb{F}_2 \otimes_{\mathbb{Z}} N) \stackrel{\text{Eq. (7)}}{=} 0.$$

Now, Proposition 3.2 implies the claim. $\diamond$

Claim 3.11. *The left $\mathbb{Q}[F]$ -module $I_{\mathbb{Q}[F]} / {}^F I_{\mathbb{Q}[\tilde{H}]}$ is $\mathcal{D}_{\mathbb{Q}[F]}$ -torsion-free. In particular,*

$$\text{tor}_{\mathbb{Q}[F]}(I_{\mathbb{Q}[F]} / {}^F I_{\mathbb{Q}[H]}) \leq {}^F I_{\mathbb{Q}[\tilde{H}]} / {}^F I_{\mathbb{Q}[H]}.$$

Proof. Put $M = I_{\mathbb{Z}[F]} / {}^F I_{\mathbb{Z}[\tilde{H}]}$. Assume that $\text{tor}_{\mathbb{Z}[F]}(M) \neq \{0\}$. Observe that M is free as a $\mathbb{Z}$ -module and that $M / \text{tor}_{\mathbb{Z}[F]}(M)$ is $\mathbb{Z}$ -torsion-free. Thus, there exists $m \in \text{tor}_{\mathbb{Z}[F]}(M)$ such that $m \notin 2M$. Put

$$\overline{M} = M / \mathbb{Z}[F]m.$$

By Eq. (1) we have

$$(8) \quad \beta_0^{\mathbb{Q}F}(\mathbb{Q} \otimes_{\mathbb{Z}} M) \leq \beta_0^{\mathbb{F}_2 F}(\mathbb{F}_2 \otimes_{\mathbb{Z}} M), \quad \text{and} \quad \beta_0^{\mathbb{Q}F}(\mathbb{Q} \otimes_{\mathbb{Z}} \overline{M}) \leq \beta_0^{\mathbb{F}_2 F}(\mathbb{F}_2 \otimes_{\mathbb{Z}} \overline{M}).$$

Since $m \in \text{tor}_{\mathbb{Z}[F]}(M)$,

$$(9) \quad \beta_0^{\mathbb{Q}F}(\mathbb{Q} \otimes_{\mathbb{Z}} M) = \beta_0^{\mathbb{Q}F}(\mathbb{Q} \otimes_{\mathbb{Z}} \overline{M}),$$

and since $m \notin 2M$ and $\mathbb{F}_2 \otimes_{\mathbb{Z}} \overline{M}$ is $\mathcal{D}_{\mathbb{F}_2[F]}$ -torsion-free,

$$(10) \quad \beta_0^{\mathbb{F}_2 F}(\mathbb{F}_2 \otimes_{\mathbb{Z}} M) = \beta_0^{\mathbb{F}_2 F}(\mathbb{F}_2 \otimes_{\mathbb{Z}} \overline{M}) + 1.$$

By Property (P3),

$$\beta_1^{\mathbb{F}_2[F]}(\mathbb{F}_2 \otimes_{\mathbb{Z}} M) = 0.$$

Thus,

$$\begin{aligned} \beta_0^{\mathbb{F}_2[F]}(\mathbb{F}_2 \otimes_{\mathbb{Z}} M) &= \chi^{\mathbb{F}_2[F]}(\mathbb{F}_2 \otimes_{\mathbb{Z}} M) = \text{rk}(F) - \text{rk}(\tilde{H}) = \\ &= \beta_0^{\mathbb{Q}F}(\mathbb{Q} \otimes_{\mathbb{Z}} M) - \beta_1^{\mathbb{Q}F}(\mathbb{Q} \otimes_{\mathbb{Z}} M) \leq \beta_0^{\mathbb{Q}F}(\mathbb{Q} \otimes_{\mathbb{Z}} M). \end{aligned}$$

Therefore, taking into account Eq. (8), we conclude that

$$(11) \quad \beta_0^{\mathbb{F}_2[F]}(\mathbb{F}_2 \otimes_{\mathbb{Z}} M) = \beta_0^{\mathbb{Q}F}(\mathbb{Q} \otimes_{\mathbb{Z}} M).$$

Hence

$$\begin{aligned} \beta_0^{\mathbb{Q}F}(\mathbb{Q} \otimes_{\mathbb{Z}} \overline{M}) &\stackrel{\text{Eq. (9)}}{=} \beta_0^{\mathbb{Q}F}(\mathbb{Q} \otimes_{\mathbb{Z}} M) \stackrel{\text{Eq. (11)}}{=} \\ &= \beta_0^{\mathbb{F}_2[F]}(\mathbb{F}_2 \otimes_{\mathbb{Z}} M) \stackrel{\text{Eq. (10)}}{=} \beta_0^{\mathbb{F}_2[F]}(\mathbb{F}_2 \otimes_{\mathbb{Z}} \overline{M}) + 1, \end{aligned}$$

that contradicts Eq. (8). Therefore, $\text{tor}_{\mathbb{Z}[F]}(M) = 0$. Hence $\mathbb{Q} \otimes_{\mathbb{Z}} M$ is $\mathcal{D}_{\mathbb{Q}[F]}$ -torsion-free. $\diamond$

3.5. Proof of Corollary 1.2. Let H be a finitely generated subgroup of F . In the introduction we have defined

$$\bar{\pi}(H \leq F) = \min\{\text{rk}(L) : H \leq L \leq F\}.$$

We obtain Corollary 1.2 as a consequence of the following result.

Theorem 3.12. *Let K be a field of characteristic 0 or 2, F a free group and H a finitely generated subgroup of F . Then*

$$(12) \quad \bar{\pi}(H \leq F) = \text{rk}(H) - \beta_1^{K[F]}(I_{K[F]}/^F I_{K[H]}).$$

Moreover, if $\tilde{H}$ is the $\mathcal{D}_{K[F]}$ -closure of H , then $\text{rk}(\tilde{H}) = \bar{\pi}(H \leq F)$.

Proof. First observe that if P is the prime subfield of a field K , then, by Proposition 2.5,

$$\beta_1^{K[F]}(I_{K[F]}/^F I_{K[H]}) = \beta_1^{P[F]}(I_{P[F]}/^F I_{P[H]}).$$

Thus, it is enough to show the theorem in the case where K is a prime field.

Notice that for a finitely generated subgroup L of F containing H , we have that

$$\begin{aligned} \text{rk}(L) &= \text{rk}(F) - \chi^{K[F]}(I_{K[F]}/^F I_{K[L]}) = \\ &= \text{rk}(F) - \beta_0^{K[F]}(I_{K[F]}/^F I_{K[L]}) + \beta_1^{K[F]}(I_{K[F]}/^F I_{K[L]}) \geq \\ &= \text{rk}(F) - \beta_0^{K[F]}(I_{K[F]}/^F I_{K[L]}) \geq \text{rk}(F) - \beta_0^{K[F]}(I_{K[F]}/^F I_{K[H]}), \end{aligned}$$

and so,

$$\begin{aligned} (13) \quad \bar{\pi}(H \leq F) &\geq \text{rk}(F) - \beta_0^{K[F]}(I_{K[F]}/^F I_{K[H]}) = \\ &= \chi^{K[F]}(I_{K[F]}/^F I_{K[H]}) + \text{rk}(H) - \beta_0^{K[F]}(I_{K[F]}/^F I_{K[H]}) = \\ &= \text{rk}(H) - \beta_1^{K[F]}(I_{K[F]}/^F I_{K[H]}). \end{aligned}$$

This implies one direction in Eq. (12).

Now we prove the result for $K = \mathbb{Q}$, and so, when K is a field of characteristic 0. A similar argument gives the theorem for fields of characteristic 2.

Let $\tilde{H}$ be the L^2 -closure of H in F . Then

$$\begin{aligned} (14) \quad \bar{\pi}(H \leq F) &\leq \text{rk}(\tilde{H}) = \text{rk}(F) - \chi^{\mathbb{Q}[F]}(I_{\mathbb{Q}[F]}/^F I_{\mathbb{Q}[\tilde{H}]}) = \\ &= \text{rk}(F) - \beta_0(I_{\mathbb{Q}[F]}/^F I_{\mathbb{Q}[\tilde{H}]}) = \text{rk}(F) - \beta_0(I_{\mathbb{Q}[F]}/^F I_{\mathbb{Q}[H]}) = \\ &= \text{rk}(H) - \beta_1^{\mathbb{Q}[F]}(I_{\mathbb{Q}[F]}/^F I_{\mathbb{Q}[H]}). \quad \square \end{aligned}$$

In the case of fields K of characteristic $p \neq 0, 2$ we do not know that free groups are $\mathcal{D}_{K[F]}$ -subgroup rigid, and, therefore, the previous proof cannot be applied in these cases.

The following corollary is a strong version of Corollary 1.2.

Corollary 3.13. *Let F be a free group and H a finitely generated subgroup. Then the following are equivalent.*

- (1) H is compressed in F ;
- (2) H is inert in F ;
- (3) H is strongly inert in F ;
- (4) H is L^2 -independent in F ;
- (5) for any field K of characteristic 0 or 2, H is $\mathcal{D}_{K[F]}$ -independent in F .

Proof. The implications $(5) \Rightarrow (4) \Rightarrow (3) \Rightarrow (2) \Rightarrow (1)$ have been discussed before.

We will prove now $(1) \Rightarrow (5)$. Let K be a field of characteristic 0 or 2. If H is compressed, then $\bar{\pi}(H \leq F) = \text{rk}(H)$. By Theorem 3.12, $\beta_1^{K[F]}(I_{K[F]}/^F I_{K[H]}) = 0$. Hence H is $\mathcal{D}_{K[F]}$ -independent in F . $\square$

3.6. Proof of Corollary 1.4. Let H be a subgroup of F . We say that H is $\mathcal{D}_{K[F]}$ -closed if

$$\text{tor}_{K[F]}(I_{K[F]}/^F I_{K[H]}) = \{0\}.$$

By Lemma 2.4, a subgroup of F is L^2 -closed if and only if it is $\mathcal{D}_{\mathbb{Q}[F]}$ -closed. In this section, we will show the following stronger version of Corollary 1.4.

Corollary 3.14. *Let F a free group and H a finitely generated subgroup of F . Then the following are equivalent.*

- (1) H is strictly compressed in F ;
- (2) H is strictly inert in F ;
- (3) H is L^2 -closed in F ;
- (4) for any field K of characteristic 0 or 2, H is $\mathcal{D}_{K[F]}$ -closed.

Proof. The implications $(2) \Rightarrow (1)$ and $(4) \Rightarrow (3)$ are clear.

Let us first show that $(3) \Rightarrow (2)$. Let H be an L^2 -closed (equivalently, $\mathcal{D}_{\mathbb{Q}[F]}$ -closed) subgroup of F , and let U be an arbitrary subgroup of F . Observe that since the left $\mathbb{Q}[F]$ -module $I_{\mathbb{Q}[F]}/^F I_{\mathbb{Q}[H]}$ is $\mathcal{D}_{\mathbb{Q}[F]}$ -torsion-free, it is also $\mathcal{D}_{\mathbb{Q}[U]}$ -torsion-free as a left $\mathbb{Q}[U]$ -module. Thus, since the left $\mathbb{Q}[U]$ -module $I_{\mathbb{Q}[U]}/^U I_{\mathbb{Q}[H \cap U]}$ is a submodule of $I_{\mathbb{Q}[F]}/^F I_{\mathbb{Q}[H]}$, $I_{\mathbb{Q}[U]}/^U I_{\mathbb{Q}[H \cap U]}$ is also $\mathcal{D}_{\mathbb{Q}[U]}$ -torsion-free.

Thus, in order to show $(3) \Rightarrow (2)$, it is enough to prove $(3) \Rightarrow (1)$. Let L be a finitely generated subgroup of F properly containing H . Then

$$\begin{aligned} \text{rk}(L) &= \text{rk}(F) - \chi^{\mathbb{Q}[F]}(I_{\mathbb{Q}[F]}/^F I_{\mathbb{Q}[L]}) = \\ &= \text{rk}(F) - \chi^{\mathbb{Q}[F]}(I_{\mathbb{Q}[F]}/^F I_{\mathbb{Q}[H]}) + \chi^{\mathbb{Q}[F]}(^F I_{\mathbb{Q}[L]}/^F I_{\mathbb{Q}[H]}) = \\ &= \text{rk}(H) + \chi^{\mathbb{Q}[F]}(^F I_{\mathbb{Q}[L]}/^F I_{\mathbb{Q}[H]}) \stackrel{\text{Proposition 2.7}}{>} \text{rk}(H). \end{aligned}$$

Now let us show $(1) \Rightarrow (4)$. Let H be a strictly compressed subgroup of F . Let $\tilde{H}$ be its $\mathcal{D}_{K[F]}$ -closure (which does not depend on the field K). If $H \neq \tilde{H}$, then we should have $\text{rk}(\tilde{H}) > \text{rk}(H)$. But this contradicts Theorem 3.12. Hence $H = \tilde{H}$. $\square$

3.7. Proof of Corollary 1.5. By Corollary 3.13, L_1 and L_2 are L^2 -independent in F . In particular they are inert. Hence $\text{rk}(L_1 \cap L_2) \leq \text{rk}(L_1)$, and so, $L_1 \cap L_2 \in \text{Crit}(H \leq F)$.

On the other hand, if $L \in \text{Crit}(H \leq F)$, then

$$\begin{aligned} \beta_0^{\mathbb{Q}[F]}(^F I_{\mathbb{Q}[L]}/^F I_{\mathbb{Q}[H]}) &= \chi^{\mathbb{Q}[F]}(^F I_{\mathbb{Q}[L]}/^F I_{\mathbb{Q}[H]}) + \beta_1^{\mathbb{Q}[F]}(^F I_{\mathbb{Q}[L]}/^F I_{\mathbb{Q}[H]}) \stackrel{\text{Lemma 2.9}}{\leq} \\ &= \text{rk}(L) - \text{rk}(H) + \beta_1^{\mathbb{Q}[F]}(^F I_{\mathbb{Q}[L]}/^F I_{\mathbb{Q}[H]}) \stackrel{\text{Theorem 3.12}}{=} 0. \end{aligned}$$

Thus, if $L = \langle L_1, L_2 \rangle$, then

$$\beta_0^{\mathbb{Q}[F]}(^F I_{\mathbb{Q}[L]}/^F I_{\mathbb{Q}[H]}) \leq \beta_0^{\mathbb{Q}[F]}(^F I_{\mathbb{Q}[L_1]}/^F I_{\mathbb{Q}[H]}) + \beta_0^{\mathbb{Q}[F]}(^F I_{\mathbb{Q}[L_2]}/^F I_{\mathbb{Q}[H]}) = 0.$$

and

$$(15) \quad \beta_1^{\mathbb{Q}[F]}(^F I_{\mathbb{Q}[L]}/^F I_{\mathbb{Q}[H]}) \stackrel{\text{Lemma 2.9}}{\geq} \beta_1^{\mathbb{Q}[F]}(^F I_{\mathbb{Q}[L_1]}/^F I_{\mathbb{Q}[H]}) = \beta_1^{\mathbb{Q}[F]}(I_{\mathbb{Q}[F]}/^F I_{\mathbb{Q}[H]}).$$

Therefore,

$$\begin{aligned} \text{rk}(L) &= \text{rk}(H) + \chi^{\mathbb{Q}[F]}({}^F I_{\mathbb{Q}[L]}/{}^F I_{\mathbb{Q}[H]}) = \\ &= \text{rk}(H) - \beta_1^{\mathbb{Q}[F]}({}^F I_{\mathbb{Q}[L]}/{}^F I_{\mathbb{Q}[H]}) \stackrel{\text{Theorem 3.12}}{=} \\ &= \bar{\pi}(H \leq F) + \beta_1^{\mathbb{Q}[F]}(I_{\mathbb{Q}[F]}/{}^F I_{\mathbb{Q}[H]}) - \beta_1^{\mathbb{Q}[F]}({}^F I_{\mathbb{Q}[L]}/{}^F I_{\mathbb{Q}[H]}) \stackrel{\text{Eq. (15)}}{\leq} \bar{\pi}(H \leq F). \end{aligned}$$

Hence $L \in \text{Crit}(H \leq F)$. $\square$

4. FINAL COMMENTS

The paper raises several questions, which we will gather in this section. Our proof of $\mathcal{D}_{\mathbb{F}_2[F]}$ -subgroup rigidity of a free group F depends on a peculiarity of the left $\mathbb{F}_2[F]$ -module $I_{\mathbb{F}_2[F]}$ (Lemma 3.6). However, we strongly believe that there are other methods that work for all primes.

Conjecture 1. *Let F be a free group and p an odd prime. Then F is $\mathcal{D}_{\mathbb{F}_p[F]}$ -subgroup rigid.*

The previous problem can be divided in two subproblems.

Conjecture 2. *Let F be a free group and p an odd prime. Let H be finitely generated subgroup of F and $\tilde{H}$ its L^2 -closure. Then*

- (1) $\beta_0^{\mathbb{F}_p[F]}(I_{\mathbb{F}_p[F]}/{}^F I_{\mathbb{F}_p[H]}) = \beta_0^{\mathbb{Q}[F]}(I_{\mathbb{Q}[F]}/{}^F I_{\mathbb{Q}[H]})$ and
- (2) *the left $\mathbb{F}_p[F]$ -module $I_{\mathbb{F}_p[F]}/{}^F I_{\mathbb{F}_p[\tilde{H}]}$ is $\mathcal{D}_{\mathbb{F}_p[F]}$ -torsion-free.*

The invariant $\bar{\pi}(H \leq F)$ is related with the **primitivity rank of H in F** introduced by Puder and Parzanchevski [PP15]:

$$\pi(H \leq F) = \min\{\text{rk}(L) : H \lhd L \leq F, H \text{ is not a free factor of } L\}.$$

It is clear that

$$\bar{\pi}(H \leq F) = \min\{\pi(H \leq F), \text{rk}(H)\}.$$

By analogy, if K is a field, we can define

$$\begin{aligned} \pi_K(H \leq F) &= \min\{\text{rk}(N) : {}^F I_{K[H]} \lhd N \leq I_{K[F]}, \\ &\quad {}^F I_{K[H]} \text{ is not a free summand of } N\}. \end{aligned}$$

The following conjecture is a variation of [EWPS24, Conjecture 1.9].

Conjecture 3. *Let F be a free group and H a finitely generated subgroup. Then for any field K , $\pi_K(H \leq F) = \pi(H \leq F)$.*

As mentioned in the introduction, we anticipate that our methods can be extended to establish L^2 -subgroup rigidity not only for free groups. To provide a specific focus, we put forth the following conjecture.

Conjecture 4. *Locally indicable groups are L^2 -subgroup rigid.*

On the other hand we expect that there are non L^2 -subgroup rigid groups.

Problem 5. *Construct non L^2 -subgroup rigid (torsion-free) groups.*

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